

Agile Sprint Planning and User Story Workshop

Smart Sports Stadium Operations Platform

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1 1. Project Vision and Stakeholders

Vision: To build an intelligent, real-time stadium operations platform that enhances spectator safety, reduces waiting times, and optimizes overall event management through data-driven decision-making and seamless coordination across all operational domains.

Objectives:

- Monitor crowd movement and density in real time.
- Optimize entry/exit flows and parking availability.
- Coordinate security operations via a centralized command center.
- Track concession performance and generate event analytics.

Key Stakeholders:

Stakeholder	Role / Interest
Spectators	Safe entry, short queues, real-time info (parking, gates).
Stadium Management	Oversee operations, ensure profitability, maintain reputation.
Security Team	Monitor threats, respond to incidents, control access.
Operations Staff	Manage parking, concessions, cleaning, logistics.
Event Organizers	Schedule events, allocate resources, coordinate with management.
IT Team	Maintain platform, networks, data security, hardware.
Concessionaires	Track sales, manage inventory, reduce queue times.

2 2. Epics and User Stories

Epic 1: Real-Time Crowd Monitoring

Epic 2: Entry/Exit Flow Optimization

Epic 3: Security Operations Coordination

Epic 4: Parking Management

Epic 5: Concession Performance Tracking

Epic 6: Event Analytics and Reporting

User Stories:

ID	User Story
US-01	As a security officer , I want to view real-time crowd density heatmaps for each stadium zone, so that I can proactively prevent overcrowding and ensure spectator safety.
US-02	As a spectator , I want to receive digital tickets with QR codes on my mobile, so that I can enter the stadium quickly without physical ticket checks.
US-03	As an operations manager , I want automated alerts when crowd density exceeds safe thresholds, so that I can dispatch staff to manage the situation immediately.
US-04	As a spectator , I want to see real-time parking availability before arriving, so that I can choose the nearest open spot and reduce traffic congestion.
US-05	As a concession manager , I want to monitor sales and inventory in real time, so that I can restock items before they run out during peak demand.
US-06	As a security chief , I want to log and track incidents from a centralized dashboard, so that all security personnel have a unified view of ongoing threats.
US-07	As a stadium manager , I want a post-event analytics report with crowd flow, sales, and incident data, so that I can improve planning for future events.
US-08	As a spectator , I want estimated waiting times displayed at each entry gate, so that I can choose the fastest queue and reduce my entry time.

3. Acceptance Criteria (Given–When–Then)

US-01 – Crowd Density Heatmaps

- **Given** CCTV cameras are operational in all zones, **When** a security officer opens the monitoring dashboard, **Then** a color-coded heatmap updating every 5 seconds is displayed for each zone.
- **Given** the system is online, **When** crowd density in any zone exceeds 80% capacity, **Then** the zone is highlighted in red with an audible alert.

US-02 – Digital Ticketing

- **Given** a spectator has purchased a ticket, **When** they open the mobile app, **Then** a scannable QR code valid for one entry is displayed.
- **Given** the QR code is scanned at a turnstile, **When** the code is valid and unused, **Then** the turnstile opens and the ticket is marked as used.

US-03 – Automated Alerts

- **Given** safe crowd thresholds are configured per zone, **When** live sensor data exceeds the threshold, **Then** an alert is pushed to the operations manager within 3 seconds.
- **Given** an alert is active, **When** the manager acknowledges it, **Then** the alert status changes to “In Progress” and logs the timestamp.

US-04 – Parking Availability

- **Given** IoT sensors are installed in parking lots, **When** a spectator opens the parking view, **Then** the count of available spots per lot is accurate within 10 seconds of change.
- **Given** a lot reaches 95% occupancy, **When** the spectator searches for parking, **Then** that lot is marked “Almost Full” and alternative lots are suggested.

US-06 – Incident Logging

- **Given** a security officer is logged in, **When** they create a new incident report, **Then** the incident appears on the chief’s dashboard within 2 seconds with GPS location.
- **Given** an incident is open, **When** the assigned officer updates the status to “Resolved”, **Then** the incident is archived and a resolution timestamp is recorded.

4 4. Prioritized Product Backlog

Rank	User Story	Epic	Points	Priority Justification
1	US-01: Crowd Density Heatmaps	Epic 1	8	Critical for safety; highest business value.
2	US-03: Automated Alerts	Epic 1	5	Directly supports safety and proactive response.
3	US-02: Digital Ticketing	Epic 2	5	Reduces entry queues; high customer impact.
4	US-06: Incident Logging	Epic 3	5	Essential for security coordination.
5	US-04: Parking Availability	Epic 4	5	Improves pre-event experience; medium value.
6	US-08: Gate Wait Times	Epic 2	3	Nice-to-have; depends on digital ticketing.
7	US-05: Concession Tracking	Epic 5	5	Revenue impact; lower safety priority.
8	US-07: Event Analytics	Epic 6	8	Post-event value; depends on other modules.

Prioritization Rationale: Safety-critical stories (crowd monitoring, alerts, security) are ranked highest. Entry optimization follows due to direct spectator impact. Parking and concessions are medium priority. Analytics, being dependent on data from other modules, is lower for early sprints.

5 5. Sprint Backlog – Sprint 1 (2 Weeks)

Sprint Goal: Deliver a foundational real-time monitoring and alert system that enables the security and operations teams to view crowd density and receive threshold alerts, ensuring basic spectator safety capabilities are operational.

Selected User Stories:

ID	User Story	Story Points
US-01	View real-time crowd density heatmaps	8
US-03	Automated alerts for crowd thresholds	5
US-02	Digital ticketing with QR codes	5
Total Sprint Capacity		18

Development Tasks Breakdown:

Story	Tasks	Owner / Notes
US-01	T1.1: Integrate CCTV feed APIs into backend.	Backend Dev
	T1.2: Implement AI crowd-density analysis module.	ML Engineer
	T1.3: Build heatmap visualization dashboard.	Frontend Dev
	T1.4: Write unit tests for density calculation.	QA Engineer
US-03	T3.1: Define configurable threshold rules per zone.	Backend Dev
	T3.2: Implement alert engine with push notifications.	Backend Dev
	T3.3: Create alert acknowledgment UI for managers.	Frontend Dev
	T3.4: End-to-end testing of alert latency (<3s).	QA Engineer
US-02	T2.1: Design ticket database schema (QR, status).	Backend Dev
	T2.2: Develop QR code generation API.	Backend Dev
	T2.3: Build mobile ticket display screen.	Mobile Dev
	T2.4: Integrate turnstile scanner hardware API.	IoT/DevOps

6. Effort Estimation (Story Points)

Technique: Planning Poker (Fibonacci: 1, 2, 3, 5, 8, 13, 21).

Baseline: A simple “Hello World” dashboard widget = 1 point.

ID	Story	Points	Estimation Justification
US-01	Crowd Density Heatmaps	8	High complexity: AI/ML integration, multi-camera feed processing, real-time visualization. Comparable to 8-point baseline of a complex data pipeline.
US-03	Automated Alerts	5	Moderate complexity: rule engine, push notifications, acknowledgment flow. Less complex than US-01 but requires reliability testing.
US-02	Digital Ticketing	5	Moderate complexity: QR generation, mobile UI, hardware integration. Well-understood domain but cross-platform work adds effort.
US-04	Parking Availability	5	Similar to US-02; IoT sensor integration is moderate effort.
US-06	Incident Logging	5	CRUD operations with real-time sync; moderate due to GPS and role-based views.
US-08	Gate Wait Times	3	Low complexity: depends on existing gate data; mainly aggregation and display.
US-05	Concession Tracking	5	POS integration and inventory logic; moderate complexity.
US-07	Event Analytics	8	High complexity: data warehouse, multi-source aggregation, report generation.

Velocity Assumption: Team velocity = 18–20 points per 2-week sprint (based on 4 developers + 1 QA).

7 7. Sprint Goal and Expected Deliverables

Sprint 1 Goal: “Establish real-time crowd visibility and basic safety alerting so that security and operations teams can monitor spectator density and respond to threshold breaches before they become dangerous.”

Expected Deliverables at Sprint End:

1. **Operational Dashboard:** A web-based dashboard displaying live crowd density heatmaps for all stadium zones, updating every 5 seconds.
2. **Alert System:** Configurable threshold rules with automatic push alerts to operations managers when zones exceed 80% capacity; alerts acknowledged within the UI.
3. **Digital Ticket MVP:** Mobile app screen showing a scannable QR code for purchased tickets; backend API for validation and single-use enforcement.
4. **Sprint Review Demo:** Live demonstration of crowd monitoring and alert features using simulated camera feeds and test data.
5. **Sprint Retrospective Report:** Documented lessons learned, velocity measurement (18 points completed), and identified improvements for Sprint 2.